

The value of x , for which the polynomials $x^2 - 1$ and $x^2 - 2x + 1$ vanish simultaneously, is

(a) 2

(b) -2

(c) -1

(d) 1

Lowest value of $x^2 + 4x + 2$ is

- | | |
|-------|----------|
| (a) 0 | (b) -2 |
| (c) 2 | (d) 4 |

If the square of difference of the zeroes of the quadratic polynomial $x^2 + px + 45$ is equal to 144, then the value of p is

(a) ± 9

(b) ± 12

(c) ± 15

(d) ± 18

If the zeroes of the quadratic polynomial $x^2 + (a + 1)x + b$ are 2 and -3 , then

(a) $a = -7, b = -1$

(b) $a = 5, b = -1$

(c) $a = 2, b = -6$

(d) $a = 0, b = -6$

Assertion : $(2 - \sqrt{3})$ is one zero of the quadratic polynomial then other zero will be $(2 + \sqrt{3})$.

Reason : Irrational zeros (roots) always occurs in pairs.

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
- (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
- (c) Assertion (A) is true but reason (R) is false.
- (d) Assertion (A) is false but reason (R) is true.

Assertion : If both zeros of the quadratic polynomial $x^2 - 2kx + 2$ are equal in magnitude but opposite in sign then value of k is $\frac{1}{2}$.

Reason : Sum of zeros of a quadratic polynomial $ax^2 + bx + c$ is $-\frac{b}{a}$

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- (d) Assertion (A) is false but reason (R) is true.

If -1 is a zero of the polynomial $f(x) = x^2 - 7x - 8$, then calculate the other zero.

Find a quadratic polynomial, the sum and product of whose zeroes are 6 and 9 respectively. Hence find the zeroes.

If α and β are the zeroes of polynomial $p(x) = 3x^2 + 2x + 1$, find the polynomial whose zeroes are $\frac{1-\alpha}{1+\alpha}$ and $\frac{1-\beta}{1+\beta}$.

Find a quadratic polynomial whose zeroes are reciprocals of the zeroes of the polynomial $f(x) = ax^2 + bx + c$, $a \neq 0$, $c \neq 0$.

The zeroes of the quadratic polynomial $x^2 + 99x + 127$ are

- (a) both positive
- (b) both negative
- (c) one positive and one negative
- (d) both equal

The zeroes of the quadratic polynomial $x^2 + kx + k$ where $k \neq 0$,

- (a) cannot both be positive
- (b) cannot both be negative
- (c) are always unequal
- (d) are always equal

Assertion : $x^3 + x$ has only one real zero.

Reason : A polynomial of n th degree must have n real zeroes.

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
- (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
- (c) Assertion (A) is true but reason (R) is false.
- (d) Assertion (A) is false but reason (R) is true.

If α and β are the zeroes of a quadratic polynomial such that $\alpha + \beta = 24$ and $\alpha - \beta = 8$. Find the quadratic polynomial having α and β as its zeroes.